AI-Powered Personalized Mobile Package Customization & Usage Management System

Degree Cohort: BSc (Honours) Data Science / Software Engineering
Module Code: PUSL3190 — Computing Individual / Group Project

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1. Executive Summary

Modern Telecommunication Service Providers (TSPs) predominantly rely on static, pre-packaged tiered plans (e.g., standard combinations of data, voice minutes, and SMS allowances). This operational model generates significant resource inefficiency: light users pay for unused allocations that expire unutilized, while heavy users experience service truncation before billing cycles complete.

This project proposes a data-driven, machine-learning-powered mobile package customization and consumption monitoring system. The platform enables consumers to dynamically build custom packages tailored to their specific needs while utilizing multivariate time-series forecasting models to predict upcoming usage trends and recommend optimized package configurations. Furthermore, the system incorporates real-time usage monitoring with non-linear alert thresholds to mitigate unexpected exhaustion of telecom allowances.

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2. Problem Statement & Research Gap

2.1 The Industry Problem

Standard telecommunication offerings enforce rigid multi-resource bundles. For example, a customer purchasing a fixed Rs. 998 package receives:

- 30 GB Data
- 1,000 Voice Minutes
- 1,000 SMS
- Access to 11 unlimited applications (30-day validity)

However, empirical consumption patterns reveal significant mismatches:

- **User Scenario A (Over-allocation / Financial Waste):** Consumes 20 GB Data, 150 Minutes, 50 SMS, and 5 Apps. A substantial portion of the purchased resource quota expires unused.

- **User Scenario B (Under-allocation / Service Disruption):** Consumes 29 GB Data, 900 Minutes, and 800 SMS within 20 days, causing early package exhaustion.

The same fixed package fails to suit both profiles efficiently.

2.2 The Research Gap

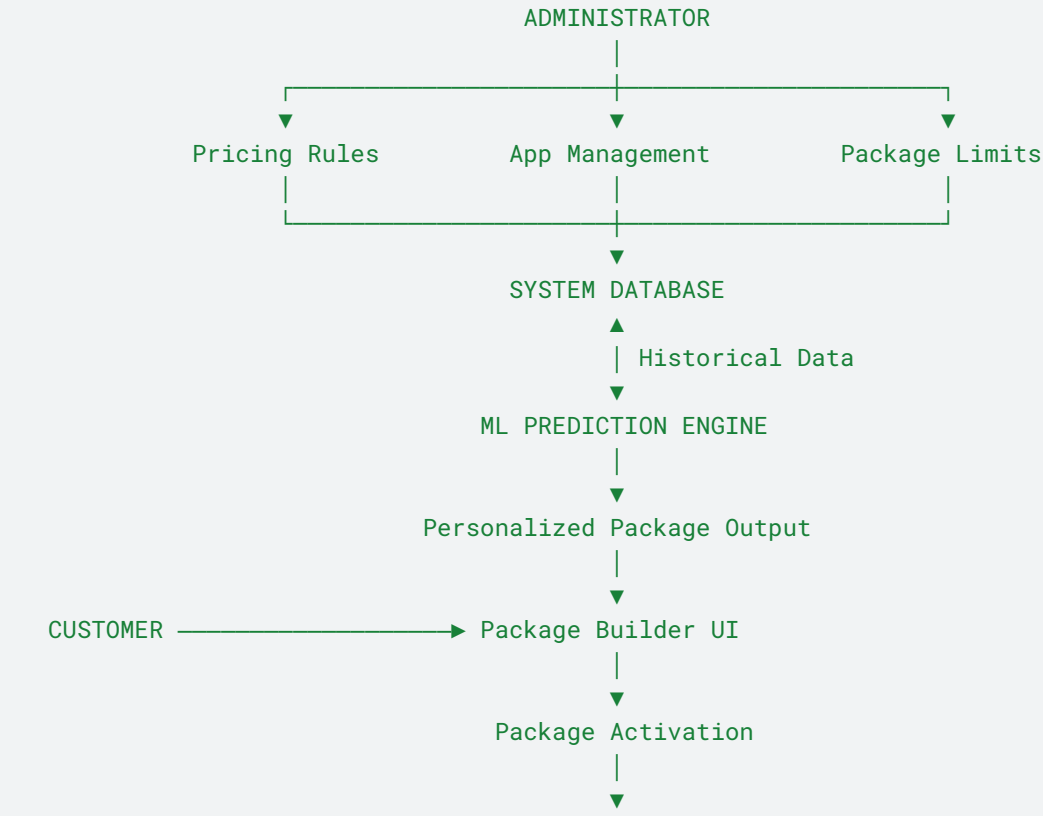
Existing commercial telecom portals feature static package selection interfaces with manual add-ons. Current academic literature and industry software lack an integrated end-to-end framework that dynamically combines:

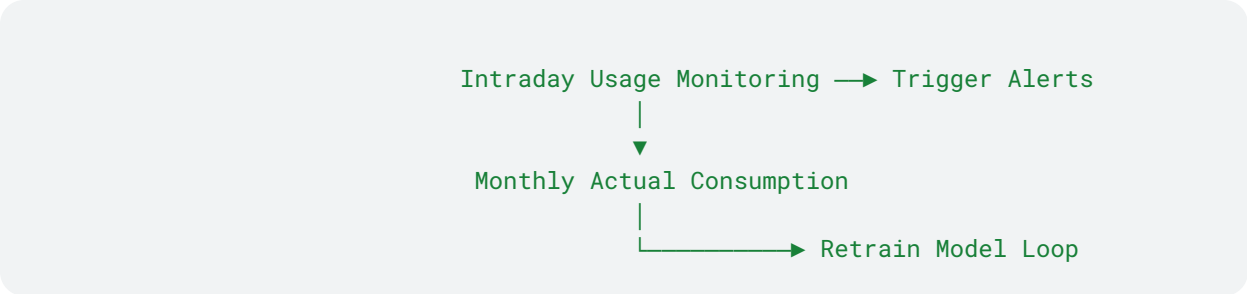
1. Flexible multi-resource pricing engines controlled via administrative rules (rather than hardcoded values).
2. Continuous machine learning prediction loops based on historical multi-resource consumption logs.
3. Intraday non-linear usage anomaly detection and threshold-based alerting.

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3. Comprehensive System Architecture

None





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4. Main System Features & Functional Modules

4.1 Customer / User Module

1. **Authentication & Profile Management:**
    - User registration, secure login, mobile number binding, telecom provider selection, profile editing.
    - View active package status and complete historical subscription logs.
2. **Interactive Package Customization Builder:**
    - Dynamic resource adjustment sliders:
        - **Data:** 5 GB ⟶ 10 GB ⟶ 15 GB ⟶ 20 GB ⟶ 25 GB ⟶ 30 GB+
        - **Calls:** 100 min ⟶ 200 min ⟶ 500 min ⟶ 1,000 min+
        - **SMS:** 50 ⟶ 100 ⟶ 500 ⟶ 1,000+
        - **Unlimited App Checkboxes:** Multi-select zero-rated apps (WhatsApp, Facebook, YouTube, Instagram, TikTok, Netflix, etc.).
    - Client-side reactive price calculation updating estimated costs instantly upon modifying any slider or app option.
3. **AI Recommendation Hub:**
    - Display predicted upcoming monthly consumption (e.g., predicted ~22 GB Data, ~200 mins, ~60 SMS, 5 Apps).
    - Options for one-click **[Accept Recommendation]** or **[Customize Package]** to retain complete customer agency.
4. **Intelligent Usage Dashboard & Analytics:**
    - Quota gauges (Remaining vs. Total Purchased).
    - Daily usage indicators comparing today's consumption against the contextual expected baseline.
    - Historical consumption charts (Data, Call, and SMS trends across previous billing cycles).

4.2 Administrator Module

1. **Dynamic Pricing Rules Engine:**
    - Administrative rule management defining rate matrices:
        - Data unit rate ($1\text{ GB} = \text{Rs. } X$)
        - Voice unit rate ($100\text{ Mins} = \text{Rs. } Y$)
        - SMS unit rate ($100\text{ SMS} = \text{Rs. } Z$)
        - App tier pricing ($\text{Per selected app} = \text{Rs. } A$)
    - System recalculates user portal prices in real time based on these dynamic database rules.
2. **Application Catalog Management:**
    - Add, edit, activate, or deactivate zero-rated apps available for package selection.
    - Assign app categories and specific app tier pricing.
3. **Usage Data & ML Diagnostics Portal:**
    - Manage anonymized user consumption records.
    - View model accuracy diagnostics (MAE, RMSE, MAPE, training timestamps, versioning).
    - Recommendation performance analytics: total recommended, accepted count, modified count, rejected count.

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5. Machine Learning & Predictive Modeling Pipeline

5.1 Problem Formulation

The system models monthly usage prediction as a **Multi-Output Time-Series Regression Task**. Given a customer's historical vector over T billing cycles:

$$X_i = \{ (\text{Data}_t, \text{Minutes}_t, \text{SMS}_t, \text{Apps}_t) \}_{t=1}^{T}$$

The objective is to forecast optimal resource requirements for cycle $T + 1$:

$$\hat{Y}_{i, T+1} = (\hat{\text{Data}}_{T+1}, \hat{\text{Minutes}}_{T+1}, \hat{\text{SMS}}_{T+1}, \hat{\text{Apps}}_{T+1})$$

5.2 Comparative Models Benchmark

The project evaluates three primary model architectures:

1. **Baseline Statistical Model:** Auto-ARIMA / Exponential Smoothing.
2. **Multivariate Machine Learning:** Multi-Output Random Forest Regressor and XGBoost Regressor.
3. **Deep Learning:** Long Short-Term Memory (LSTM) Recurrent Neural Network for temporal dependencies.

5.3 Continuous Feedback & Model Retraining Cycle

1. Historical usage feeds the prediction engine.
2. Prediction generates an automated recommendation.
3. Customer accepts, modifies, or declines the package.
4. Actual usage during the active month is tracked and recorded.
5. New usage metrics append to the database to retrain and fine-tune models periodically.

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6. Daily Usage Monitoring & Non-Linear Alert Engine

Rather than using simple linear averages (`$\text{Total GB} / 30 \text{ Days}$`), the monitoring engine applies non-linear baseline curves based on historical user behavior (e.g., weekend spikes vs. weekday low usage):

- **Weekday Baseline Allocation:** Lower daily baseline (e.g., `$600\text{ MB/day}$`).
- **Weekend Baseline Allocation:** Higher daily baseline (e.g., `$1.2\text{ GB/day}$`).

Alert Threshold Escalation Matrix

Status Level	Condition Trigger	Alert Message / Action
Normal (Green)	`$\text{Actual Usage} \le 105\%$` of Expected Baseline	✅ "Your usage is within your expected daily range."
Warning (Yellow)	`$105\% < \text{Actual Usage} \le 130\%$` of Expected Baseline	⚠️ "Your data usage is higher than your expected daily baseline."
High Risk (Red)	`$\text{Actual Usage} > 130\%$` or Projected Expiry < 5 Days	🔴 "Critical: Your consumption rate will exhaust your quota before package expiration."

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7. Data Schema & Engineering Requirements

To train models and support real-time monitoring, each customer billing record contains:

Field Name	Type	Description
`customer_id`	String	Unique identifier
`billing_cycle`	Date	Start date of package cycle
`package_type`	Enum	Custom / Recommended / Fixed
`data_allowance_gb`	Float	Purchased data limit
`data_used_gb`	Float	Actual data consumed
`call_allowance_min`	Integer	Purchased voice minutes
`calls_used_min`	Integer	Actual voice minutes used
`sms_allowance`	Integer	Purchased SMS count
`sms_used`	Integer	Actual SMS used
`apps_selected_count`	Integer	Number of unlimited apps chosen
`apps_used_count`	Integer	Apps actively utilized
`package_price`	Decimal	Calculated total price paid
`cycle_duration_days`	Integer	Package validity period (e.g., 30)

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8. Primary Research Questions (RQs)

- **RQ1:** *How accurately can multivariate machine learning models forecast an individual user's monthly data, voice, and text requirements based on historical usage logs?*
- **RQ2:** *To what extent does a personalized package recommendation system reduce resource redundancy (unused purchased allocations) compared to fixed telecom bundles?*
- **RQ3:** *How effective are contextual daily usage monitoring baselines in detecting potential quota exhaustion before package expiry?*

- **RQ4:** *How do comparative machine learning models (Random Forest vs. XGBoost vs. LSTM) evaluate in terms of prediction error (MAE, RMSE) and computational runtime overhead?*

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9. End-to-End Customer Journey (Demonstration Scenario)

1. **Login:** Customer logs into the portal and views their active plan status.
2. **Customize:** Customer navigates to the *Package Builder* and adjusts resources (e.g., lowering data from 30 GB to 22 GB, minutes from 1,000 to 200, SMS to 100, apps to 5).
3. **Live Price:** Price dynamically recalculates on screen from the admin pricing engine rules.
4. **Activation:** Customer activates the customized plan.
5. **Intraday Monitoring:** Customer uses 1.2 GB on a weekday (exceeding the 600 MB baseline).
6. **Alert Issued:** System issues a Warning alert on the dashboard.
7. **Cycle End & Store:** At cycle end, actual consumption metrics are stored in the database.
8. **AI Recommendation:** The ML model runs on the updated history and generates the recommended package for next month.
9. **Next Cycle:** Customer logs in, receives the AI recommendation, accepts or modifies it, and repeats the cycle.

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10. Core Technical Description Statement

> "This project proposes an AI-powered personalized mobile package customization system where customers can dynamically tailor data, call, SMS, and app benefits with dynamically calculated pricing, while the system learns from their historical usage patterns to recommend their next package and continuously monitors intraday consumption to provide intelligent usage alerts. An admin portal manages packages, pricing rules, application catalogs, usage datasets, and recommendation analytics."